

HIMANSHU SHELKE

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Research Summary

Integrated Master's student (BS-MS) bridging Mathematics, Computer Science, and Machine Learning. Specialized in numerical linear algebra, continuous optimization, and structural matrix decomposition. Proven ability to bypass standard high-level libraries to engineer custom first- and second-order optimization algorithms and extreme data compression pipelines, directly applicable to reducing computational memory costs in large-scale machine learning models.

Education

Indian Institute of Science Education and Research (IISER), Thiruvananthapuram Aug 2023 – May 2028

Integrated Master of Science (BS-MS), Data Science

GATE 2026: Data Science and Artificial Intelligence

Feb 2026

Successfully prepared for and completed the national-level examination.

Technical Competencies

Mathematics & Optimization: Numerical linear algebra, custom matrix factorizations (Compact SVD, QR), continuous/non-convex optimization algorithms, applied calculus.

Computer Science & Engineering: Python, C/C++, algorithmic time-complexity analysis, high-performance computational pipelines, hardware-constrained execution.

Machine Learning: Feature space pruning (L1 Regularization, Correlation Filtering), PyTorch, extreme data compression, continuous-action Reinforcement Learning.

Academic Projects & Experience

High-Dimensional Matrix Reconstruction (Compact SVD)

Feb 2026 – Mar 2026

Numerical Linear Algebra & Compression

- Implemented explicit Compact SVD protocols from scratch for robust matrix reconstruction, bypassing standard Partial SVD libraries to ensure rigorous mathematical handling of highly correlated data.
- Strictly isolated and ignored zero eigenvalues to preserve the underlying data topology and prevent structural corruption during extreme dimensional reduction and compression.

Hardware-Constrained Edge Intelligence & Feature Pruning

Aug 2026

Memory & Computational Cost Reduction

- Developed a machine learning pipeline specifically optimized to reduce memory and computational costs for execution on constrained edge hardware.
- Executed severe feature selection methodologies, utilizing L1 regularization and correlation filtering to aggressively prune the high-dimensional data space.
- Mathematically traded a marginal 2% reduction in absolute accuracy for guaranteed, low-latency execution speeds within strict hardware parameters.

Stochastic Environments & Continuous Optimization

Mar 2026 – Apr 2026

Custom Algorithmic State Updates

- Implemented Proximal Policy Optimization (PPO) in a custom PyTorch environment, mathematically stabilizing extreme variance in highly volatile continuous spaces.
- Bypassed standard high-level abstractions to explicitly write custom Conjugate Gradient descent algorithms, maximizing computational efficiency during network state updates.

Algorithmic Benchmarking & Scientific Reporting

Apr 2026

Experimental Validation & Academic Publication

- Executed an experimental validation of deterministic search algorithms against model-free reinforcement learning architectures.
- Evaluated computational models to establish strict convergence rates and Big-O time complexities.
- Drafted a comprehensive research paper describing the results in a formal IEEE-formatted LaTeX document, demonstrating readiness for high-level academic conference submissions (e.g., NeurIPS, ICML).

Participation Certificate

for the
Summer School on
Mathematical Finance

I have arranged a summer school from 3rd June to 24th July 2026 at IISER Pune. It is my pleasure to confirm that Mr. Himanshu Satish Shelke from IISER Thiruvananthapuram attended all sessions of the summer school. This academic event aimed to teach fundamental concepts in price process modelling, fair pricing of derivatives, and portfolio optimization.

During the summer school, he learnt many concepts and actively participated in the group discussions. He has also presented his own investigation to others.



30th July 2026, IISER Pune

Dr. Anindya Goswami
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